AgriApp Firestore Schema v1 (SmartAgri+)

> Scope: Offline-first, multilingual, AI-driven farm assistant with soil inference (grids/Bhuvan/IoT), hyperlocal weather, market-demand pricing, visual disease assistance, and a decision engine for “next crop”.

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1) Collections Overview (high-level)

- `users/{uid}` — farmer profile, prefs, devices, top-level pointers.
- `users/{uid}/lands/{landId}` — each parcel/field owned or managed by the user.
- `users/{uid}/lands/{landId}/soilSnapshots/{snapshotId}` — timestamped soil readings (from IoT, Bhuvan, grids, lab/manual).
- `users/{uid}/lands/{landId}/sensors/{sensorId}` — registered field sensors.
- `users/{uid}/lands/{landId}/crops/{cropId}` — both **past** and **current** crops for that land.
- `users/{uid}/alerts/{alertId}` — price/weather/disease alerts.
- `users/{uid}/conversations/{convId}` — AI chat sessions.
- `users/{uid}/conversations/{convId}/messages/{msgId}` — per-message logs.
- `aiDecisions/{decisionId}` — persisted “decision runs” for explainability & audit.
- `marketSnapshots/{docId}` — normalized price & demand by region+commodity+date.
- `commodities/{commodityId}` — canonical crop catalog, sowing windows, fert guides.
- `weatherDaily/{regionId}/days/{yyyymmdd}` — daily hyperlocal weather cache.
- `diseaseReports/{reportId}` — user-submitted photo diagnoses and outcomes.
- `iotSensors/{sensorId}` — global registry for devices (also mirrored under user/land).
- `iotSensors/{sensorId}/readings/{tsId}` — raw timeseries (optional if using soilSnapshots only).
- `contentArticles/{articleId}` — multilingual KB/how-to content.
- `translations/{langCode}/keys/{key}` — UI + system string cache (optional).
- `notifications/{notifId}` — system fan-out queue (Cloud Functions target).
- `auditLogs/{logId}` — optional compliance trail.

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2) Documents & Fields (detailed)

2.1 `users/{uid}`

```
{
  "name": "string",
  "phone": "string",
  "city": "string",
  "village": "string",
  "pincode": "string",
  "preferred_language": "ta-IN | hi-IN | en-IN | ...",
```

```
  "voice": { "preferred": true, "tts_voice": "string", "stt_engine":
"bhashini" },
  "profile_photo_url": "gs://...",
  "createdAt": "timestamp",
  "updatedAt": "timestamp",
  "last_login": "timestamp",
  "location_last": { "_lat": 0, "_long": 0 },
  "fcm_tokens": ["string"],
  "offline_mode": true,
  "consents": { "data_share": true, "research": false },
  "role": "farmer | expert | admin"
}
```

2.2 `users/{uid}/lands/{landId}`

```
{
  "name": "North Plot",
  "area_acres": 2.5,
  "centroid": { "_lat": 11.34, "_long": 77.72 },
  "boundaries": [ {"_lat": 11.3401, "_long": 77.7201}, ... ],
  "address": {"village":"","tehsil":"","district":"","state":"","pincode":""},
  "soil_type": "Alluvial | Black | Red | Laterite | ...",
  "irrigation_type": "Canal | Borewell | Rainfed | Drip",
  "cropping_system": "Mono | Inter | Mixed | Rotation",
  "ownership_docs": ["gs://..."],
  "soil_profile_latest": "soilSnapshotId-or-null",
  "createdAt": "timestamp",
  "updatedAt": "timestamp"
}
```

2.3 `users/{uid}/lands/{landId}/soilSnapshots/{snapshotId}`

```
{
  "source": "iot | bhuvan | soilgrid | lab | manual",
  "provenance": {
    "provider": "NRSC-Bhuvan | ISRIC-SoilGrids | deviceId | labId",
    "tile_id": "string-or-null",
    "resolution_m": 250,
    "depth_cm": [0,30],
    "raw_payload_url": "gs://..."
  },
  "measures": {
    "pH": 6.7,
    "moisture_pct": 21.3,
    "EC_dS_m": 0.8,
```

```
    "organic_carbon_pct": 0.7,
    "N_kg_ha": 250,
    "P_kg_ha": 55,
    "K_kg_ha": 180,
    "micronutrients": {"Zn": 0.9, "B": 0.4}
  },
  "weather_context": { "regionId": "IN-TN-ERD", "day": "2025-08-31", "rain_mm": 12, "tmax_c": 33, "tmin_c": 24 },
  "model_inference": {
    "recommended_nutrients": [
      {"name":"Urea","amount_kg_ha": 45, "timing":"basal"},
      {"name":"DAP","amount_kg_ha": 30, "timing":"top-dress"}
    ],
    "next_crop_candidates": [
      {"commodityId":"paddy", "fit":0.82, "rationale":"soil pH ~ neutral; water availability; market"},
      {"commodityId":"maize", "fit":0.71}
    ],
    "risk_score": 0.28,
    "model_version": "soil-ai@2025.08"
  },
  "takenAt": "timestamp",
  "createdAt": "timestamp"
}
```

2.4 `users/{uid}/lands/{landId}/sensors/{sensorId}`

```
{
  "type": "soilProbe | weatherNode | gateway",
  "make": "string",
  "installDate": "timestamp",
  "lastSeenAt": "timestamp",
  "battery_pct": 86,
  "firmware": "1.2.3",
  "mqtt_topic": "farm/{uid}/{landId}/{sensorId}",
  "status": "active | dormant | retired"
}
```

2.5 `users/{uid}/lands/{landId}/crops/{cropId}` (Past + Current)

```
{
  "crop_name": "Paddy",
  "variety": "ADT-45",
  "season": "Kharif | Rabi | Zaid",
  "status": "current | past",
```

```
  "sowing_date": "timestamp",
  "expected_harvest_month": 12,
  "expected_harvest_year": 2025,
  "harvest_date_actual": "timestamp-or-null",
  "area_used_acres": 1.8,
  "invested_inr": 48000,
  "earned_inr": 76000,
  "sold": true,
  "yield_kg": 2900,
  "market_channel": "domestic | export",
  "fertilizer_plan": [ {"name":"Urea","amount_kg_ha":45,"date":"timestamp"} ],
  "price_context": {
    "domestic": {"avg_inr_per_quintal": 2200, "demand_index": 0.64},
    "export": {"avg_inr_per_quintal": 2600, "demand_index": 0.51}
  },
  "notes": "string",
  "createdAt": "timestamp",
  "updatedAt": "timestamp"
}
```

2.6 `users/{uid}/alerts/{alertId}`

```
{
  "type": "weather | disease | price | task | irrigation",
  "message": "string",
  "severity": "info | warn | critical",
  "related": {"landId":"","cropId":"","commodityId":""},
  "lang": "ta-IN",
  "tts_audio_url": "gs://...",
  "createdAt": "timestamp",
  "readAt": "timestamp-or-null"
}
```

2.7 `users/{uid}/conversations/{convId}` and `/messages/{msgId}`

```
// conversation
{
  "mode": "chat | voice | voice+image",
  "lang": "ta-IN",
  "persona": "agri-assistant",
  "startedAt": "timestamp",
  "endedAt": "timestamp",
  "context": {"landId":"optional","cropId":"optional"}
}
```

```
// message
{
  "role": "user | assistant | tool",
  "text": "string",
  "audio_url": "gs://...",
  "image_urls": ["gs://..."],
  "detected_lang": "ta-IN",
  "translated_text_en": "string",
  "tool_calls": [{"name":"soilRecommend","args":{}}],
  "createdAt": "timestamp"
}
```

2.8 `aiDecisions/{decisionId}` (global)

```
{
  "uid": "string",
  "landId": "string",
  "inputs": {
    "soilSnapshotId": "string",
    "weatherRegionId": "string",
    "market_window": {"from":"2025-08-01","to":"2025-12-31"},
    "budget_inr": 60000,
    "constraints": {"water":"limited","duration_days_max":120}
  },
  "output": {
    "recommended_crop": {"commodityId":"paddy","variety":"ADT-45"},
    "expected_duration_days": 115,
    "expected_cost_inr": 52000,
    "expected_yield_kg": 3000,
    "gross_return_inr": 78000,
    "net_return_inr": 26000,
    "fertilizer_plan": [ ... ],
    "explainability": {"top_factors":[{"feature":"market_price","weight":0.41}]}
  },
  "rationale": "string",
  "model_version": "next-crop@2025.08",
  "createdAt": "timestamp"
}
```

2.9 `marketSnapshots/{docId}` (composite key suggestion: `${regionId}_${commodityId}_${yyyymmdd}`)

```
{
  "regionId": "IN-TN-ERD",
  "commodityId": "paddy",
```

```
  "date": "2025-08-31",
  "source": "agmarknet | apis",
  "unit": "INR/quintal",
  "min": 2100,
  "max": 2400,
  "avg": 2250,
  "demand_index": 0.62,
  "export_window": {"open": true, "notes": "nearby port demand"},
  "updatedAt": "timestamp"
}
```

2.10 `commodities/{commodityId}`

```
{
  "name": "Paddy",
  "categories": ["Cereal"],
  "varieties": ["ADT-45","CO-51"],
  "growing_days": {"min":100, "max":130},
  "sowing_windows": [
    {"regionId":"IN-TN","start":"06-15","end":"08-15"}
  ],
  "fertilizer_guides": [ {"stage":"basal","urea_kg_ha":50}, ... ],
  "disease_catalog_refs": ["blast","bacterial_leaf_blight"],
  "images": ["gs://..."],
  "createdAt": "timestamp",
  "updatedAt": "timestamp"
}
```

2.11 `weatherDaily/{regionId}/days/{yyyymmdd}`

```
{
  "regionId": "IN-TN-ERD",
  "date": "2025-08-31",
  "forecast": {"tmax_c":34, "tmin_c":24, "rain_mm":8, "humidity_pct": 82,
"wind_kph": 12},
  "alerts": [{"type":"thunderstorm","severity":"warn","window":"18:00-23:00"}],
  "source": "OWM | IMD",
  "generatedAt": "timestamp"
}
```

2.12 `diseaseReports/{reportId}`

```
{
  "uid": "string",
```

```
  "landId": "string",
  "crop_name": "Paddy",
  "image_urls": ["gs://..."],
  "model_output": {"label":"Rice Blast","confidence":0.91},
  "recommendations": ["Use Tricyclazole 0.6g/L", "Avoid late evening
irrigation"],
  "follow_up_required": true,
  "createdAt": "timestamp",
  "resolvedAt": "timestamp-or-null"
}
```

2.13 `iotSensors/{sensorId}` & `/readings/{tsId}` (optional global mirror)

```
// iotSensors/{sensorId}
{
  "uid": "owner uid",
  "landId": "string",
  "type": "soilProbe",
  "registeredAt": "timestamp",
  "secret_hash": "string",
  "status": "active"
}

// readings/{tsId}
{
  "ts": "timestamp",
  "pH": 6.8,
  "moisture_pct": 19.5,
  "temp_c": 27.2
}
```

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3) Cloud Storage (paths)

- `gs://app-bucket/profiles/{uid}/profile.jpg`
- `gs://app-bucket/lands/{uid}/{landId}/docs/{filename}`
- `gs://app-bucket/soil/raw/{uid}/{landId}/{snapshotId}.json`
- `gs://app-bucket/disease/{reportId}/{imageName}.jpg`
- `gs://app-bucket/tts/{uid}/{alertId}.mp3`

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4) Indexing Plan (Firestore Composite Indexes)

- `marketSnapshots`: `(commodityId ASC, regionId ASC, date DESC)`